

Biomet Microfixation Temporomandibular Joint Replacement System: a 3-year follow-up study of patients treated during 1995 to 2005.

PMID: 22449430 | Indexed in PubMed

To assess the safety and efficacy of a stock alloplastic total temporomandibular joint (TMJ) implant system, the Biomet Microfixation TMJ Replacement System. During a 10-year multicenter clinical trial from 1995 to 2005, 442 Biomet Microfixation TMJ Replacement Systems were implanted in 288 patients (154 bilaterally and 134 unilaterally). Patients were followed at landmark times, including the date of surgery and at 1 month, 3 months, 1 year, 1 year 6 months, and 3 years. The 3 major metrics that were evaluated were preoperative and postoperative pain, interference with eating, and maximal incisal opening. Paired t tests and comparison analyses were used to assess outcomes. There was statistically significant improvement in pain level ($P = .0001$), jaw function ($P = .0001$), and incisal opening ($P = .0001$). Although there were complications necessitating the removal of 14 of 442 implants (3.2%), there were no device-related mechanical failures. The clinical study presented supports the conclusion that a stock TMJ alloplastic replacement, based on sound orthopedic and biomedical principles, is a safe and efficacious option when alloplastic reconstruction of the TMJ is indicated.

Improved outcomes after alloplastic TMJ replacement: analysis of a multicenter study from Australia and New Zealand.

PMID: 24820776 | Indexed in PubMed

The aim of the present study was to assess pain, function, diet, and quality of life (QOL) in subjects undergoing alloplastic temporomandibular joint (TMJ) reconstruction. A total of 72 subjects were sent a questionnaire an average of 4 years after undergoing either unilateral or bilateral total joint replacement with a custom TMJ Concepts or stock Biomet total joint replacement device. The changes in pain, function, diet, and quality of life were assessed. Of the 72 subjects, 52 returned the questionnaire. Pain, function, diet, and QOL were all significantly improved at a mean follow-up of 46.3 months. The complications reported included an altered facial sensation and facial weakness. No implant failures were reported. The placement of an alloplastic TMJ total joint replacement can provide benefits for patients in the areas of pain, function, diet, and QOL. Additional research is required in this area.

Predictability and Feasibility of Total Alloplastic Temporomandibular Joint Reconstruction using DARSN TM Joint Prosthesis for patients in Indian subcontinent-A prospective clinical study.

PMID: 31255829 | Indexed in PubMed

The aim of this study was to investigate the feasibility of a custom made alloplastic Temporomandibular Joint (TMJ) device design in patients undergoing temporomandibular (TM) Total Joint Reconstruction (TJR). TMJ disease with functional and anatomic distortion dictates the need for TJR. There are various materials to reconstruct a TMJ. However, various factors, such as cost, availability of prosthetic joint, limit its use to tertiary health care center. Hence, we have investigated the feasibility and efficacy of the custom made alloplastic TMJ prosthesis (DARSN TM Joint Prosthesis) with the advantage of being acceptable financially

and the overall Quality of life (QoL) diagnosed with TMJ ankylosis and End Stage Joint Disease (ESJD) selected from the study population. The study group comprised of 20 patients with TMJ ankylosis or End Stage Joint Disease (ESJD) who needed TM TJR of which few subjects in the study population had history of failed previous surgery to the TMJ region. The patients underwent resection of the joint followed by TJR using the custom made alloplastic TMJ prosthesis. Various subjective and objective variables were evaluated such as the Jaw Function (JF), Inter-incisal opening (IO), Diet intake (DI), Quality of Life (QoL) using a Psychometric Modified Likert Scale and nutritional status of the patient using the Mid-Upper Arm Circumference (MUAC) as reference. All the subjective and objective variables showed significant improvement in the postoperative period as compared to the preoperative period. The JF score increased with a mean score of 6.25 ($P < 0.00001$). Postoperative mean DI score was 3.15 ($P < 0.00001$) and IO increased up to 29-38mm in 95% of the study population. The study population exhibited an improved overall QoL and nutritional status post-operatively. Follow up period of 1 year showed significant functional improvement among the study population. The results shows that the custom made alloplastic joint replacement is safe and effective and reliable alternative to treat patients with TMJ disease which restricts the normal function to a greater degree requiring TM TJR.

Pseudogout of the temporomandibular joint: immediate reconstruction with total joint arthroplasty.

PMID: 19340868 | Indexed in PubMed

Pseudogout of the temporomandibular joint (TMJ) is a rare disorder with <40 reported cases in the English-language literature. It is usually treated with surgical resection and condylectomy alone. We report the case of a 68-year-old woman presenting with a preauricular mass resulting from calcium pyrophosphate deposition (pseudogout) in the TMJ, who underwent surgical resection with immediate reconstruction. Open surgical resection to completely remove the mass confirmed substantial erosion of the condyle and disc, necessitating condylectomy and discectomy. The joint was thereafter immediately reconstructed by total joint arthroplasty using a Christensen TMJ implant sized to the patient's anatomy. The patient was able to resume a regular diet and demonstrated marked improvement in her interincisal opening. Immediate reconstruction of the TMJ with total joint arthroplasty is a safe and effective measure following removal of this (and potentially other) TMJ lesions.

Status of alloplastic total temporomandibular joint replacement procedures performed by members of the American Society of Temporomandibular Joint Surgeons.

PMID: 27567049 | Indexed in PubMed

The purpose of this study was to assess the number of alloplastic total temporomandibular joint replacement (TMJ TJR) devices implanted and the complications encountered by members of the American Society of Temporomandibular Joint Surgeons (ASTMJS). A questionnaire was developed using REDCap (Chicago, IL) and an on-line link was e-mailed four times over a 6-week period (February to March 2015) to all members of the ASTMJS. The questionnaire included eight questions related to the surgeon's TMJ TJR experience. A Likert scale was included to assess the respondents' opinions concerning the future demand for TMJ TJR and their thoughts on potential device improvements. The study sample was composed of 36 ASTMJS respondents (42.4%). TMJ TJR is performed successfully and in relatively

high numbers by members of the American Society of Temporomandibular Joint Surgeons.

Functioning of the Masticatory System in Patients with an Alloplastic Total Temporomandibular Joint Prostheses Compared with Healthy Individuals: A Pilot Study.

PMID: 36556438 | Indexed in PubMed

Most patients with temporomandibular joint (TMJ) issues are successfully treated with nonsurgical methods. However, when end-stage TMJ pathologies occur, invasive management can be required, such as a total TMJ replacement. This cross-sectional pilot study aimed to provide insight into the functioning of the masticatory system, pain, and patient satisfaction in patients treated with a total joint replacement (TJR). A cross-sectional pilot study was conducted to determine the postoperative clinical results of an alloplastic TJR TMJ. Masticatory performance and also insight into maximum voluntary bite force (MVBF), active and passive maximum mouth opening (aMMO/pMMO), pain, and patient satisfaction were measured. Masticatory performance, MVBF, and aMMO of patients with a TJR TMJ were compared with healthy individuals. Masticatory performance is equal between patients with a TJR TMJ and healthy individuals, but both MVBF and aMMO were significantly smaller in patients with a TJR TMJ. However, patients had almost no pain and were very satisfied with the TJR TMJ treatment. This study revealed that most patients with an alloplastic TJR TMJ were able to function without pain, showed good masticatory performance, and were highly satisfied with their alloplastic TJR TMJ. However, MVBF and aMMO were lower than in healthy individuals.

Temporomandibular joint reconstruction.

PMID: 27348350 | Indexed in PubMed

Temporomandibular joint (TMJ) reconstruction represents one of the more challenging clinical problems that a head and neck surgeon encounters. Fortunately, the problem is fairly uncommon; however, at the same time, clear clinical guidelines have not been formulated. The goal of this review is to present the established solutions to this difficult reconstructive challenge and highlight key developments that have been recently published. Modern-day approaches to TMJ reconstruction include a handful of surgical modalities: costochondral grafting, revascularized tissue transfer, distraction osteogenesis, and alloplastic temporomandibular joint replacement (APTMJR). Over the past several years, publications have focused primarily on distraction osteogenesis and APTMJR. In particular, APTMJR is emerging as a highly successful and versatile surgical modality when faced with TMJ defects. With long-term follow-up after APTMJR approaching 20 years at several of the pioneering centers, alloplastic total joint replacement is emerging as the clinical standard when faced with TMJ defects in a variety of clinical situations. Unfortunately, the dogma of serious complications that occurred during early development may still be stalling widespread acceptance of APTMJR. Due to the complexity of the surgery, its relative infrequency and the lack of exposure in the majority of residency training programs, this surgery is currently best performed by tertiary referral centers.

Modification of an extended total temporomandibular joint replacement (eTMJR)

classification system.

PMID: 35618505 | Indexed in PubMed

The aims of this paper were to validate a modification of an extended total temporomandibular joint replacement (eTMJR) classification system and develop a classification schematic for ease of reference. High-volume TMJ surgeons were asked to score 20 separate eTMJR devices using the updated classification system, and inter-rater variability was calculated. Using the modified classification system developed, a Conger's kappa (κ) coefficient of 0.53 was returned, suggesting moderate to good levels of agreement. The final classification system was then developed in a series of standardised graphic illustrations as visual representations of the different subcategories of eTMJR devices.

Previously Reported Approach Now Used for Temporomandibular Joint Replacement.

PMID: 32649564 | Indexed in PubMed

Over time, different techniques have been described to obtain broad access to the temporomandibular joint (TMJ), in order to allow good visibility, minimizing post-operative complications at the same time. Most of these techniques have been very useful to perform joint replacement using custom made prostheses, varying in its extension, aesthetics and functional results. The aim of this study was to present the authors' experience using the Blair approach modified by Fernández in 2015, to replace failed TMJ prostheses, when distortion of the surgical reference points has occurred.

Total Temporomandibular Joint Replacement: Stick to Stock or Optimization by Customization?

PMID: 32642034 | Indexed in PubMed

This article aims to compare the difference in postoperative results in patients treated with either a patient-specific (PSI) or a stock temporomandibular total joint replacement system. The investigators performed a systematic review concerning postoperative results after placement of either a stock total joint replacement system or a PSI. PubMed Central, Web of Science, Cochrane Library Plus, Wiley Online Library, and EMBASE were used to conduct this search. All articles up to August 15, 2018, were scrutinized. All included articles were nonrandomized cohort studies. Maximal mouth opening (MMO) and Visual Analog Scale (VAS) scores for pain and diet before and after surgery were evaluated. The Methodological Index for NonRandomized Studies scale was used for quality assessment. Weighted mean difference was calculated and pooled by meta-analysis using random-effect models. The search identified 1581 articles, of which 15 were included. The average risk of bias was low. Both systems achieved significant increases in MMO and decreased VAS pain scores at 1, 2, and 3 years after surgery. No significant difference was found between the system types. Both achieved significant improvements in dietary VAS scores, with a more significant improvement for stock implants. Due to the lack of detailed diagnostic evaluation tools allowing proper start-point categorization, there is a significant risk for selection bias in the pooled data. The PSI is more frequently chosen for cases with more significant joint degeneration, skewing postoperative results. A patient-fitted implant can provide significant operative and patient-centered advantages over a stock implant, which will likely be confirmed when observational cohort studies have included indications like the ones for stock prostheses. Furthermore, while current US

Food and Drug Administration-approved stock implants contain cobalt -chromium -molybdenum, the newly manufactured PSI are made of titanium alloy, diminishing the risks of morbidity and implant failure.
